YILDIZ TECHNICAL UNIVERSITY

Department Of Mechatronic Engineering

MKT3434 Machine Learning

HW2

**Instructor**

Ertuğrul Bayraktar

Emirhan AK 210A020

**İstanbul 2025**

# Introduction

This project extends a machine learning GUI with the integration of several key methods, including PCA (with explained variance plotting and CSV export), t-SNE and UMAP (with configurable dimensionality and interactive 2D/3D visualization), LDA for supervised reduction, K-Means clustering with inertia and silhouette score metrics, the Elbow Method for optimal k selection, and K-Fold Cross-Validation with mean and standard deviation reporting. Additionally, UMAP and t-SNE results can be exported, and all components support user-adjustable parameters from the interface.

## Principal Component Analysis (PCA)

PCA is a linear dimensionality reduction technique that projects high-dimensional data into a lower-dimensional space by maximizing variance along orthogonal axes. It is commonly used for data compression, noise reduction, and visualization.

### GUI Integration

- The GUI allows the user to select the number of principal components (n_components) via a spinbox.
- A cumulative explained variance plot is generated to visualize how much information is retained.

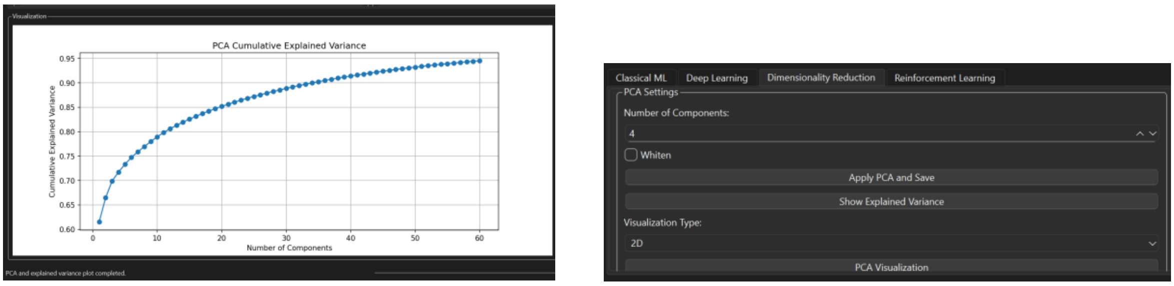



*Figure 1,2 : Cumulative Explained Variance plot of Arduino Sensor Values and PCA Group*

### PCA Visualization

After reduction, the first two(three for 3D)  principal components are used to visualize the dataset in 2D,3D.

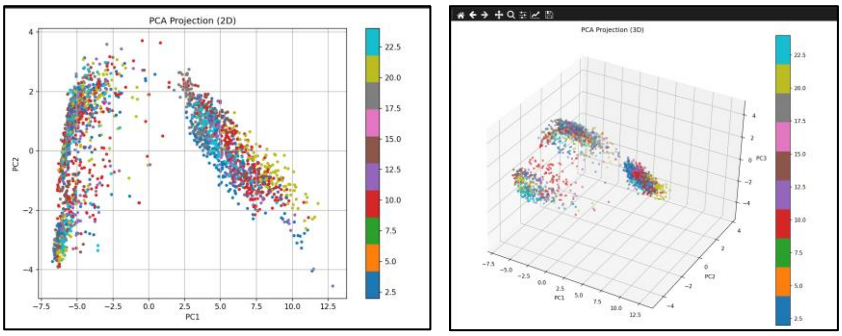
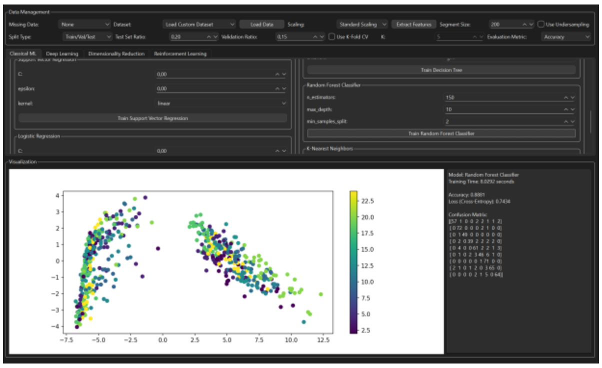
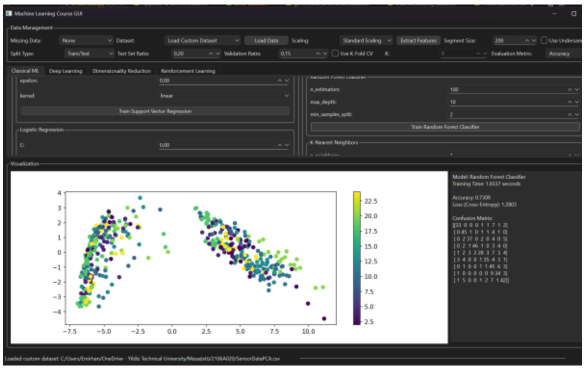



Figure 3,4 = PCA visualization 2D,3D

### Sensor Data Compression via PCA

I also added the option to save the PCA applied version as CSV. To evaluate the impact of PCA on real sensor data (e.g., EMG), two separate training runs were conducted:





✅ **With PCA:**

❌ **Without PCA:**

| Metric | With PCA | Without PCA |
|---|---|---|
| Input Feature Size | 40 components | 561 raw features |
| Training Time | **1.03 seconds** | **8.02 seconds** |
| Accuracy | **73.09%** | : **88.81%** |

Although the accuracy decreased slightly when using PCA, the training time was nearly halved and the feature dimensionality was significantly reduced. This demonstrates that PCA offers a valuable trade-off for sensor data pipelines, where reducing computational cost is often more critical than achieving the absolute highest accuracy.

## t-SNE – Nonlinear Dimensionality Reduction & Visualization

The GUI includes full support for **t-SNE**, a nonlinear technique used to visualize high-dimensional data in 2D or 3D space. The user can configure:

- perplexity (controls local vs global structure)
- n_components (2D or 3D)

Once the transformation is applied:

- An interactive **Plotly scatter plot** is generated
- **Silhouette score** is computed and shown in the chart title


Figure 7: t-SNE settings Group

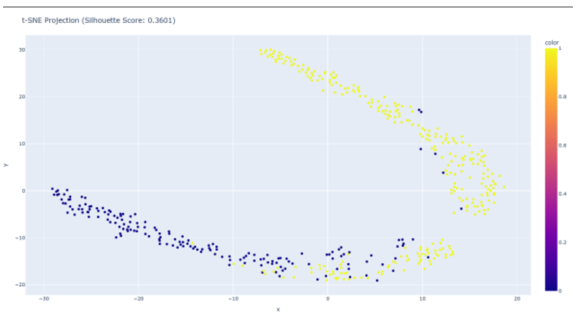



Figure 8 : t-SNE Projection

# UMAP – Flexible Dimensionality Reduction (Visual & Model-Ready)

For Faster Alternative to tsne

UMAP offers both visualization and data compression capabilities:

- Parameters n_neighbors, min_dist, and n_components are fully configurable
- The user can apply UMAP and:
  - **Visualize results in 2D or 3D**
  - **Export the reduced feature set as CSV**

This makes UMAP suitable for use both in exploratory data analysis and as input to machine learning models.

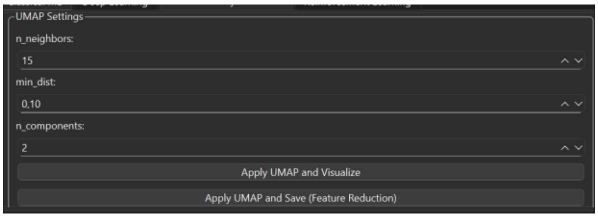



Figure 9: UMAP group

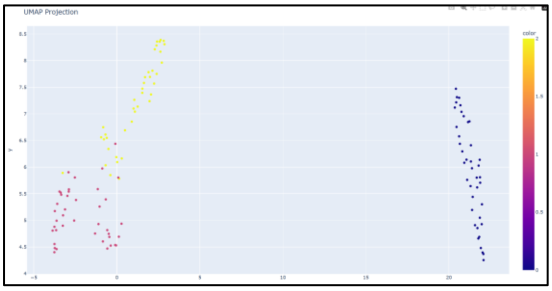



Figure 10: Umap Projection

**LDA – Supervised Dimensionality Reduction**

**Linear Discriminant Analysis (LDA)** was added as a supervised method that uses class labels to find projection axes that maximize inter-class separation.

In the GUI:

- LDA is applied in 2D (up to n_classes - 1 components)
- An interactive scatter plot is generated with class-colored clusters
- Silhouette score is calculated and displayed

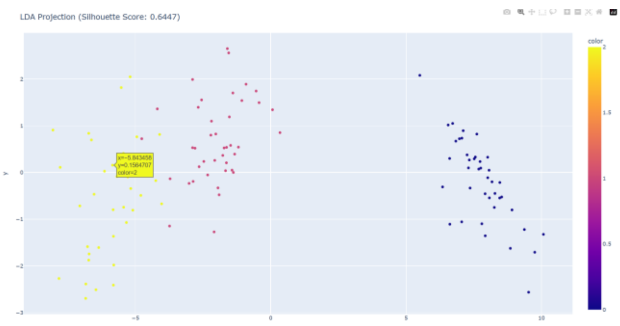



Figure11: LDA Projection

### K-Means Clustering + Evaluation Metrics

K-Means clustering is fully integrated with:

- Configurable n_clusters, n_init, and max_iter
- Clustering is visualized in 2D using PCA projection
- Both **inertia** (compactness) and **silhouette score** (cluster separation) are computed and shown

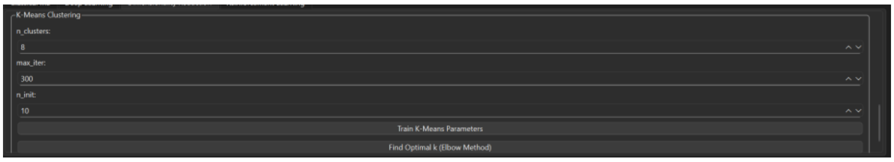



Figure 12: K-Means Group

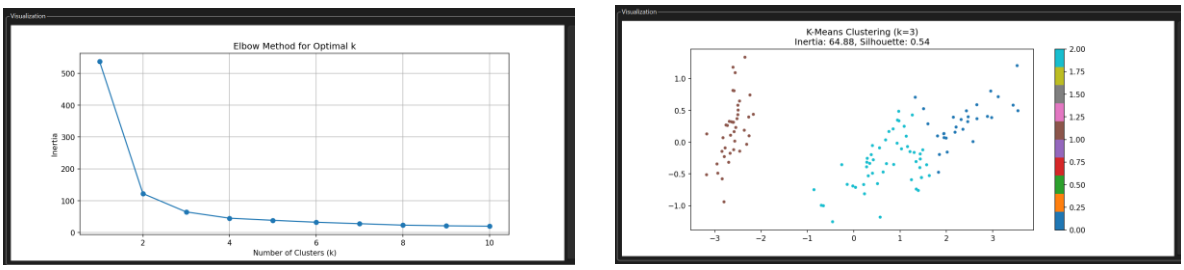



Figure 13,14: Elbow method for optimal k and with optimal k K-Means Clustering

Based on the Elbow Method plot, k = 3 was selected as the optimal cluster number, as it represents the point where the rate of inertia decrease significantly slows down.

### K-Fold Cross-Validation – Robust Model Evaluation

To ensure robust performance estimation:

- The user selects the number of folds (k)
- Metrics supported include: **Accuracy**, **MSE**, **RMSE**, **MAE**
- The GUI displays both:
    - **Mean score across folds**
    - **Standard deviation** (for performance consistency)

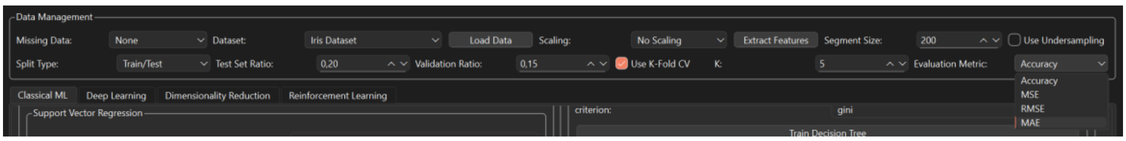



Figure 15: K-Fold parameters and check button added to Data Management

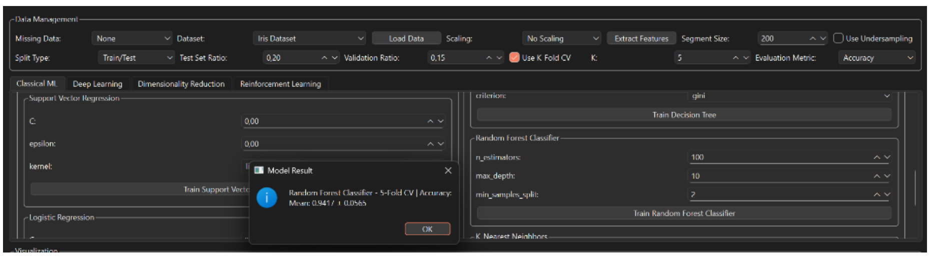



Figure 16: k-fold applied to Iris Dataset

## Raw Signal Feature Extraction + Undersampling

The system supports working with time-domain signals (e.g., EMG data):

- Data is segmented based on a configurable segment size
- Statistical features are extracted per segment:
    - mean, std, min, max, rms( EMG data has 8 channel if we extract the feature, 8x5= 40 feature)
- **Undersampling** can be applied to balance the dataset before segmentation
- The resulting feature set is saved as a CSV file

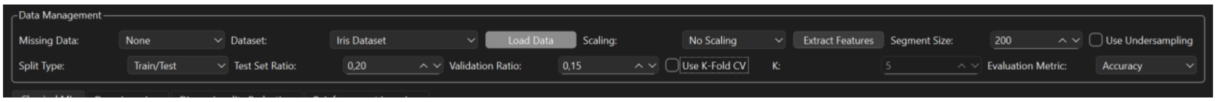



Figure 17: when click the extract features we select raw data. After that dataset with extracting feature can be saved

```
1  "time","channel1","channel2","channel3","channel4","channel5","channel6","channel7","channel8","class","label"
2  1,1e-05,-2e-05,-1e-05,-3e-05,0,-1e-05,0,-1e-05,0,"1"
3  5,1e-05,-2e-05,-1e-05,-3e-05,0,-1e-05,0,-1e-05,0,"1"
4  6,-1e-05,1e-05,2e-05,0,1e-05,-2e-05,-1e-05,1e-05,0,"1"
```

Figure 18: raw data

```
1  channel1_mean,channel1_std,channel1_min,channel1_max,channel1_rms,channel2_mean,channel2_std,channel2_min,channel2_max,channel2
2  2.9499999999999995e-05,0.00012379317428679176,-0.00022,0.0004,0.00012725957724273642,-1.8e-05,5.608921464952063e-05,-0.00021,4e
3  -3.5000000000000012e-06,0.00012264888911033806,-0.00047,0.00022,0.00012269881825021788,-5.500000000000001e-06,3.81411850890870
```

Figure 19: Extracted feature data

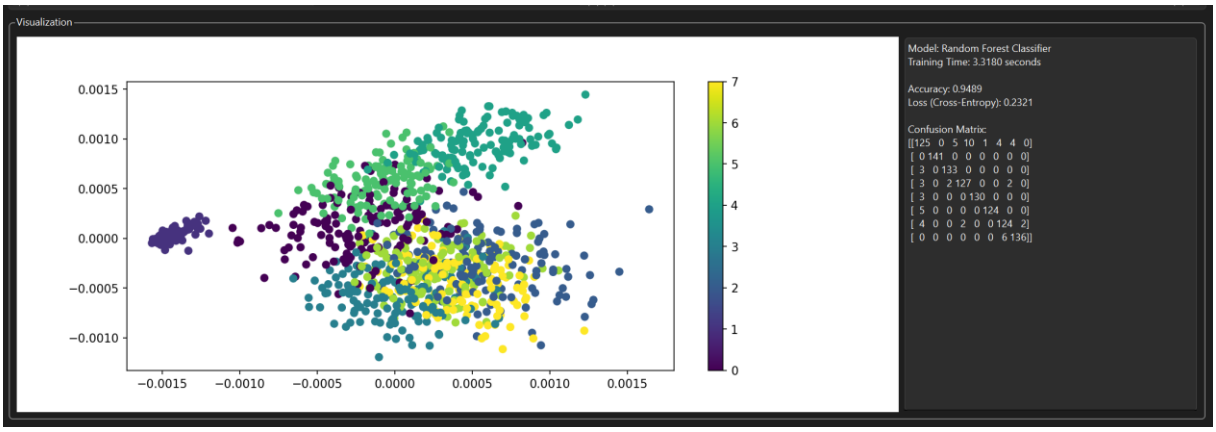
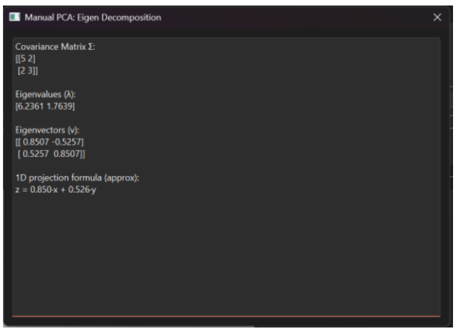
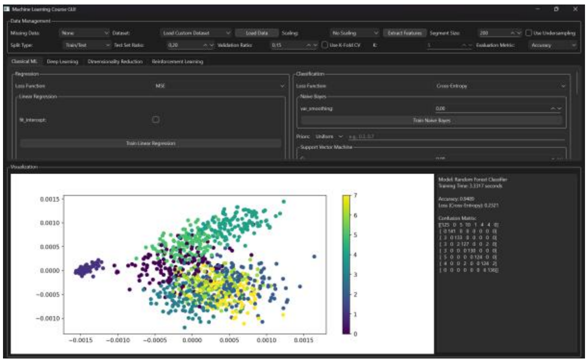


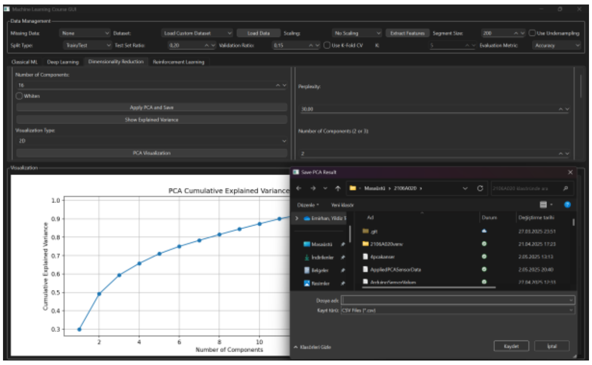

Figure 20: EMG data Train completed with accuracy %94.89

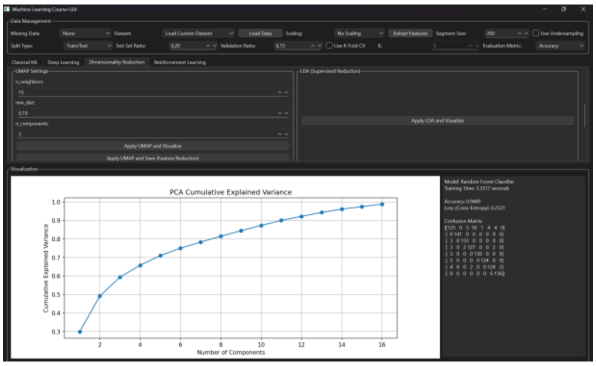
### Manual Eigen Decomposition for 1D Projection

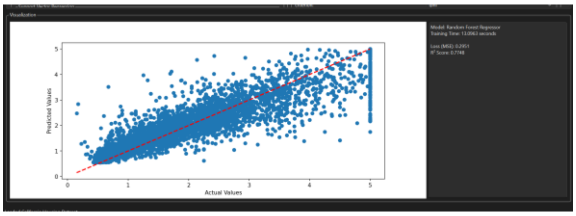


## GUI Screenshots









**Additionally, a Random Forest Regressor was integrated into the GUI. Compared to other regression models, it produced significantly higher $R^2$ scores**